SYSTEM MANIFESTO: NIDAAN.AI

FEDERATED MULTI-NATIONAL CLINICAL CORE & OPERATIONS ENGINE

Executive Summary & Core Philosophy

Nidaan.AI is a zero-downtime, offline-first, multi-country health stack engineered to connect Primary Healthcare Centers (PHC) and Community Healthcare Centers (CHC) with regional command centers and national health identity frameworks (such as India's ABDM / ABHA stack). The platform is governed by three non-negotiable architectural principles:

1. **Zero Cross-Border PHI Leaks:** Patient Identifiers (PII/PHI) are strictly isolated within local regional nodes. Central telemetry streams receive cryptographically anonymized data bundles.
2. **Clinical Safety First (Human-in-the-Loop CDSS):** AI and rules engines provide standardized draft recommendations, but licensed treating physicians retain sole sign-off authority.
3. **Resilience & Offline Autonomy:** All clinical check-ins, stock deductions, and bed updates function seamlessly offline, reconciling upon reconnection using Vector Clocks and Last-Write-Wins (LWW) conflict resolution.

1. Target Stakeholders & Dedicated User Terminals

Target Stakeholder	Dedicated Interface Terminal	Core System Objective
Patients & Citizens	Public Patient Portal	Transparent directory access, live bed availability, diagnostic search & ABHA registration.
Receptionists & Clerks	Staff Intake Terminal	Rapid walk-in registration, national identity (ABHA) verification & queue assignment.
Attending Physicians	Doctor Clinical Terminal	Consultations, ICD-11 coding, WHO EML CDSS recommendations, dosage checks & digital sign-off.
Nurses & Ward Staff	Nurse Ward Terminal	Real-time bed occupancy tracking, admissions/discharges & bio-fenced QR check-ins.
Pharmacists	Pharmacy Stock Terminal	Automatic stock deduction on doctor sign-off, inventory alerts & depletion forecasting.
District Admins & Ministry	Ministry Command Portal	Multi-facility monitoring, attendance audits, Gemini AI forecasting & anonymized analytics.

2. Complete Feature & Technical Capability Catalog

- **Haversine Bio-Fencing Telemetry:** Computes great-circle distance (R = 6,371,000m) to enforce staff presence within a strict 100.0-meter radius during shift check-ins.
- **Dynamic QR Shift Verification:** Generates cryptographic QR tokens bound to specific facilities and staff IDs to prevent proxy attendance.
- **ABDM / ABHA National Health Stack (Module A):** Verifies national identity (India ABDM M1/M2/M3) at reception with automatic profile pre-fill and local intake fallback for un-enrolled patients.

• **HL7 FHIR R4 Standard Data Model:** Transforms intake and clinical records into HL7 FHIR R4 Patient, Condition (ICD-11), and MedicationRequest schemas.

• **Zero-Trust PII Privacy Shield:** Strips names, phone numbers, exact addresses, and hashes national IDs before sending telemetry to analytical pipelines.

• **Rules-Backed CDSS Engine (Module B):** Provides WHO EML & ICD-11 grounded treatment guidelines for Malaria (1F40), Hypertension (BA00), Respiratory Infection (CA00), and Gastroenteritis (1A40).

• **CRITICAL_DOSAGE_OVERFLOW_WARNING Fail-Safe:** Bounding checker validating daily intake against max daily safety limits (e.g. Paracetamol > 4g/day). Blocks issuance unless explicit doctor override reason is provided.

• **Physician Digital Sign-off:** Mandatory doctor digital signature protocol required to commit prescriptions to pharmacy logs and trigger stock deductions.

• **Global i18n Localization Engine (Module C):** Dynamic runtime language switcher supporting en-US, hi-IN (Hindi), pt-BR (Portuguese), and ar-SA (Arabic with native RTL layout).

• **Vector Clock & LWW Sync Manager (Module D):** Offline queue ingestion resolving IndexedDB/SQLite local transaction conflicts via Vector Clock sequences and Last-Write-Wins timestamps.

• **Automated Pharmacy Stock Deduction:** Signed prescriptions automatically deduct stock levels in the live database and flag minimum safety threshold alerts.

• **Gemini AI & Velocity Stock Forecasting:** Combines local consumption velocity mathematics with Google Gemini AI (gemini-2.5-flash) to compute depletion runways and reorder warnings.

• **Live Bed Occupancy & Visual Degradation Tracking:** Monitors bed availability across General, ICU, and Emergency wards while displaying facility operational degradation badges.

3. Security, Privacy & Data Sovereignty Protocols

Role-Based Access Control (RBAC): Strictly enforces headers (X-Active-Role / X-User-Role) to restrict privileged actions (prescribing, bed allocation, inventory edits) to authorized staff.
Regional Data Sovereignty: Raw patient identifiers (PII/PHI) never leave regional storage nodes. Telemetry streams receive anonymized FHIR bundles with masked names (ANONYMIZED_PATIENT) and coarsened birth years.